# Containers over a base — state of the program

A status memo

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*Corresponds to work-in-progress commit `53db0f7`; detailed write-ups: `containers-over-a-base.pdf` (the approaches survey, with the three closedness theorems) and `admissibility-composition-product.pdf` (the $\pi_0$ characterisation) in `scotmacbeth/work-in-progress`.*

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Neil — a fast catch-up on where the "containers over different categories" program stands. The theory has crystallised into a clean two-paper package plus one live frontier. Sections 1–2 are the settled map and ledger; §3–§4 are the admissibility spine and the novelty gate; **§5 is the open-questions section you asked for and is the one to read**; §6 corrects one point in the thread you were replying to. I keep proofs to a one-line mechanism throughout — everything below is written up in full in the two PDFs.

## 1 The question, and the map

A *container* in a category $\mathcal{C}$ is a pair (shapes, positions): a set $S$ of shapes together with a position object $P_s \in \mathcal{C}$ for each $s$, extending to the endofunctor $[\![S, P]\!]X = \coprod_s [P_s, X]$. Over **Set** this is Abbott–Altenkirch–Ghani [1]. The moment one asks for "a container over a base $\mathcal{C} \neq$ **Set**", the notion forks into three-or-four inequivalent constructions:

(1) **External families**, $\mathrm{Fam}(\mathcal{C}^{\mathrm{op}})$: an *external* **Set**-indexed coproduct of internal homs, $[\![S, P]\!] = \coprod_s [P_s, -]$. This is the only one that reaches additive bases such as **Vec** (Dorta–Jarvis–Niu's $\Sigma\Pi\mathcal{C}$ construction [4]).

(2) **Indexed / dependent-polynomial** $\Sigma\Pi\Delta$ over a locally cartesian closed (LCC) base — Gambino–Kock [6].

(3) **Fibrational / comprehension**: positions as a fibration over shapes; subsumes (1) and (2) as the family and codomain fibrations.

(4) **Walker's locally-subcartesian-closed** bases [12]: an affine base-change $\nabla_f \dashv \boxtimes_f$ (subpullbacks) replacing the top adjoint when the base is not LCC.

**The organising insight.** These are not four rival definitions but a *totally ordered tower of weakenings* of the $\Sigma\Pi\Delta$ machinery, refining the local-cartesian-closure axis:

$$\underbrace{\text{LCCC}}_{\text{G–K [6]}} \supset \underbrace{\text{pullbacks}}_{\text{Weber [13]}} \supset \underbrace{\text{subpullbacks}}_{\text{Walker [12]}} \supset \text{spans.}$$

LCC asks pullback to have a right adjoint globally; Weber's distributivity pullbacks weaken this to a per-morphism (exponentiable-leg) condition; Walker weakens it further to an affine adjoint; at the bottom sit bare spans with no base-change adjoint at all.

**Where Vec falls.** $\mathrm{Fam}(\mathbf{Vec}^{\mathrm{op}})$ falls off the *bottom* of the tower: **Vec** fails every rung, always for the same two reasons. It is **non-extensive** — the comparison $A \amalg B \to A \oplus B$ from the disjoint union of underlying sets to the biproduct misses every genuine linear combination, so $\coprod^{\mathbf{Set}} \subsetneq \oplus$ (coproduct $\neq$ biproduct). And it has **no internal** $\Pi$ — it is not LCC, exponentiability fails. So *only* approach (1) can even form its semantics over **Vec**. This is the honest reason the linear world needs the external construction: not a preference, a forced move. (Von Glehn [11], TAC 33 (2018), identifies approach (2) with the codomain fibration and (1) with the family fibration; the two diverge exactly on non-extensive / non-LCC bases.) The two axes — extensivity and local cartesian closure — are the genuine coordinates of the whole landscape, and **Vec**, failing both, is the discriminating example.

## 2 Proved results: a ledger

All four statements below are proved in full in `containers-over-a-base.pdf`. I give each as a theorem plus its one-line mechanism. Write $[\![-]\!]\colon \mathrm{Fam}(\mathcal{C}^{\mathrm{op}}) \to [\mathcal{C}, \mathcal{C}]$ for the extension.

**Theorem 1** (T1, flagship — fullness is unit-connectedness)**.** $[\![-]\!]$ *is* fully faithful $\iff$ *the monoidal unit $I$ is* connected*, i.e. $\mathcal{C}(I, -)\colon \mathcal{C} \to \mathbf{Set}$ preserves small coproducts.*

*Mechanism:* the action on homs is $\prod_s$ of the comparison $\gamma\colon \coprod_t \mathcal{C}(I, X_t) \to \mathcal{C}(I, \coprod_t X_t)$, and connectedness is exactly "$\gamma$ always bijective." *This is not extensivity.* $\mathbf{Set}\times\mathbf{Set}$ is lextensive yet $[\![-]\!]$ is not full: its unit $(1,1)$ is disconnected, so for source $(\{s\}, (1,1))$ and target $(\{t_1, t_2\}, ((1,1),(1,1)))$ there are 2 container morphisms against 4 natural transformations — the two "crossed" maps are unrealised. This corrects the folklore that read "extensivity" off the **Set** case (where base and value category coincide, fusing two different theorems — ours, and Diers' reconstruction [3] which uses extensivity of the *codomain*).

**Theorem 2** (T2 — Dirichlet $\otimes$ closedness)**.** $\otimes$ *on* $\mathrm{Fam}(\mathcal{C}^{\mathrm{op}})$ *is closed* $\iff$ *the functor* $\Phi(Z) = \prod_t \coprod_r \mathcal{C}(M_r, Z \otimes Q_t)$ *is familially representable.*

*Mechanism:* a right adjoint at $(T, Q), (R, M)$ represents a functor whose restriction to the one-shape generators is $\Phi$; representability transfers through the free coproduct completion. Closed on cartesian-closed bases (recovering the **Poly** Dirichlet hom, Niu–Spivak Ex. 4.78 [10]) and on $\mathrm{Fam}_{\mathrm{fin}}(\mathbf{Vec}_{\mathrm{fd}}^{\mathrm{op}})$ *only*; it **fails** over both $\mathrm{Fam}(\mathbf{Vec}_{\mathrm{fd}}^{\mathrm{op}})$ and $\mathrm{Fam}(\mathbf{Vec}^{\mathrm{op}})$ by *dual* mechanisms — infinitely many shapes break product-closure, infinite-dimensional positions break single-factor representability. The single load-bearing condition is simultaneous *dualizability-and-summability*, met over **Vec** exactly on the finite/finite-dimensional corner.

**Theorem 3** (T4-left — $\lhd$ left-closedness is collapse)**.** *The substitution product $\lhd$ is* left-*closed* $\iff$ *it* collapses*: positions tiny (= dualizable = finite-dimensional over* **Vec***) forces $\lhd = \otimes$, symmetric.*

*Mechanism:* tininess turns $[Z, \coprod_t B_t]$ from the branching $\coprod_{\tau\colon Z \to T} \prod_d B_{\tau(d)}$ (which makes the **Set** closure non-polynomial, hence absent) into a genuine coproduct, whence $\lhd = \otimes$. Holds on $\mathrm{Fam}_{\mathrm{fin}}(\mathbf{Vec}_{\mathrm{fd}}^{\mathrm{op}})$. **The crown inversion:** extensivity, the property whose *failure* costs T1 and T2, is here the *obstruction* — non-extensivity is the *repair*. Extensivity $\perp$ $\lhd$-left-closedness. This is the survey's most surprising structural fact and is why the classification is genuinely two-dimensional rather than a single "goodness" scale.

**Theorem 4** (Approach (3) anchored — the logic of containers)**.** $\mathbf{Cont}(\mathrm{cod}) = \mathrm{Fam}(\mathrm{cod}^{\mathrm{op}})$ *is a* bifibration *and a* co-hyperdoctrine*. The fibre over $(S, \{P_s\})$ is $\prod_s (\mathbf{Set}/P_s)^{\mathrm{op}}$; the quantifiers along the shape-collapse are the* $\mathsf{All}/\mathsf{Exists}$ *predicate liftings (*$\mathsf{All} = \Pi$, $\mathsf{Exists} = \Sigma$*).*

*Mechanism / dualisation:* the container hyperdoctrine is the *fibrewise opposite* of the standard **Set**-family hyperdoctrine — container-$\exists$ is $\mathsf{All}$, container-$\forall$ is $\mathsf{Exists}$, $\wedge \leftrightarrow \vee$, and each fibre is a *co-topos*. Consequently joint Beck–Chevalley / Frobenius is *one-sided*: the right-adjoint quantifiers carry it, the left-adjoint ones fail, obstructed by co-topos non-distributivity. (Fibrational folklore — Fam preserves fibrations [7], cod is a bifibration [8]; the fibrewise-op presentation is von Glehn [11].)

## 3 The admissibility program (the spine of the generality question)

This is where your "for which bases does it generalise?" question actually lives. Say $\mathcal{C}$ is $\lhd$-*admissible* if $\mathrm{Fam}(\mathcal{C}^{\mathrm{op}})$ carries the composition product at all: for all $p, q$ there is $p \lhd q$ with $[\![p \lhd q]\!] \cong [\![p]\!] \circ [\![q]\!]$. Per-object this is *absorptivity*: every object $X$ has $X \mapsto [P, [\![q]\!]X]$ a coproduct of representables. The whole story is in `admissibility-composition-product.pdf`.

**Trichotomy.** The space of bases splits into three regimes.

- **Extensive pole.** admissible $\Rightarrow$ unit connected $\Rightarrow$ $[\![-]\!]$ fully faithful and $(-) \lhd q$ has a *left adjoint for every* $q$. One comparison map $\gamma$ powers both fullness and the adjoint — "the same lemma applied twice."
- **Collapse pole.** admissible with unit $\cong 0$ (every object copower-tiny, $\lhd = \otimes$); $(-) \lhd q$ has a left adjoint iff $|T| = 1$.
- **Inadmissible.** **Set** $\times$ **Set**, $\mathbf{Set}_*$ (pointed sets), $\mathrm{Gl}((-)^2)$.

**Extensive-CCC pole — completely characterised (the headline).**

**Theorem 5** ($\pi_0$-characterisation)**.** *On a nontrivial infinitary-lextensive cartesian-closed base,* $\mathrm{Fam}(\mathcal{C}^{\mathrm{op}})$ *is $\lhd$-admissible $\iff$ the connected-components functor $\pi_0$ preserves finite products.*

This single criterion (Lemma S: the shape object $[P, T \cdot \mathbf{1}]$ must be a copower of $\mathbf{1}$) *unifies the two inadmissible extensive bases into one phenomenon*: **Set** $\times$ **Set** fails because its unit is disconnected ($\pi_0(X \times Y) = X_1Y_1 \sqcup X_2Y_2 \neq (X_1 \sqcup X_2) \times (Y_1 \sqcup Y_2)$), and $\mathrm{Gl}((-)^2)$ fails despite a *connected* unit because $\pi_0(K \times K) = 2 \neq 1$ for the single edge $K$ (Weichsel's theorem on the graph tensor product [15]; Gl is a Grothendieck topos [9, 2]).

**Connectedness is necessary but NOT sufficient.** $\mathrm{Gl}((-)^2)$ is the witness: a topos, a connected unit, still inadmissible. So "connected unit" is strictly weaker than admissibility, and the gap is exactly the external-vs-internal distributive law (internal composition of Gambino–Kock polynomials holds in *every* topos; the external-shape law is what $\pi_0$-multiplicativity controls).

**Gap 1 is inhabited.** The region (admissible $\wedge$ non-collapse $\wedge$ non-cartesian) is non-empty: $\mathbf{Set} \times \mathbf{Vec}_{\mathrm{fd}}$ is $\lhd$-admissible on its natural locus, with an explicit $\lhd$ and a natural isomorphism — the rigid **Set** factor forces the shape set $\coprod_s T^{A_s}$, the flexible $\mathbf{Vec}_{\mathrm{fd}}$ factor absorbs it by biproduct collapse. Its unit is *disconnected*, so the extensive-pole extraction (admissible $\Rightarrow$ connected) genuinely needs its hypotheses: admissibility alone does not force connectedness. The extensive pole is a *sufficient* condition for one-bit rigidity, not a characterisation of it.

## 4 Novelty gate

Dorta–Jarvis–Niu (arXiv:2305.05655, [4]) build a $\otimes$ over a general base that matches mine exactly under the identification $\Sigma\Pi\mathcal{C} = \mathrm{Fam}((\Pi\mathcal{C})^{\mathrm{op}})$. But $\Pi\mathcal{C}$ is the *coproduct-free* product completion, so the infinitary-lextensive-CCC hypotheses of the admissibility theorems *never hold there* — DJN is therefore neither prior art for, nor a counterexample to, the admissibility results. They prove no fullness/faithfulness result and pose the base-condition question explicitly as future work ([4, §6]). MacBeth's delta is: fullness (T1), closedness (T2/T4), and change-of-enrichment (T3). *One caveat, flagged:* their composition product is *weighted* and coincides with mine only at $\mathcal{C} = 1$ (their direction object multiplies in an outer factor $u_{i,a}$ that mine omits) — the general comparison of $\triangleleft_{\mathrm{DJN}}$ with $\triangleleft$ is open.

## 5 Open questions (the part you asked for)

### THE FRONTIER — Conjecture 6.2, the absorptive dichotomy

**Conjecture 1** (Absorptive dichotomy)**.** *Every absorptive object is a tensor of a* copower-tiny *(flexible) part and a* distributive *(rigid) part. Hence every "nice" admissible base = (rigid-extensive) × (flexible-additive), and there is* ***no irreducible Gap-1 inhabitant****.*

A scoping result reduces the *whole* conjecture to a single question:

**Question 1.** Is full (infinite-dimensional) **Vec** $\triangleleft$-admissible?

The reduction: any additive base is non-extensive, so it has no rigid factor; if it were admissible it would be an *irreducible* Gap-1 inhabitant, i.e. a counterexample to 6.2. So 6.2 holds $\iff$ full **Vec** is *inadmissible*. This in turn reduces to a homological question in the additive functor category $\mathrm{Add}(\mathbf{Vec}, \mathbf{Vec})$:

**Question 2.** Is $F = \mathrm{Hom}(k^{(\mathbb{N})}, \bigoplus_{\mathbb{N}} -) = \prod_{\mathbb{N}}(\bigoplus_{\mathbb{N}} -)$ *projective* in $\mathrm{Add}(\mathbf{Vec}, \mathbf{Vec})$ — equivalently, a coproduct of representables?

**Status: OPEN.** What is settled this cycle ("computed"), and what I want you to know before spending time on it:

- The naive "$\infty$-copower obstruction" — that $F$ fails to preserve coproducts/products, hence cannot be a coproduct of representables — is **not rigorous**. Every candidate separator turns out to be a *non-obstruction*: a coproduct of representables with infinite-dimensional summands *also* fails to preserve infinite coproducts/products. Preservation is simply the wrong diagnostic here.
- A Yoneda finite-row-support rigidity lemma (**Lemma R**) is *proved*, but it kills only one candidate summand, not the dichotomy.
- The Baer–Specker / slenderness engine that refutes the **Ab**-analogue **does not transport**: over a field every vector space is free, so there are no slender objects to run the argument on. (Cited in prose only — see flag in the report.)
- A rigorous resolution needs $\mathrm{Ext}^1(F, -) \neq 0$ in $\mathrm{Add}(\mathbf{Vec}, \mathbf{Vec})$: honest functor-category homological algebra over a field. This is the single highest-value open problem in the program.

*Honest current bet:* direction A (full **Vec** inadmissible $\Rightarrow$ dichotomy holds), by analogy only — no proof.

**Secondary open questions**

- **Wildcard: nuclear / topological vector spaces** (where tiny $\neq$ finite-dimensional) — the one corner the extensive-pole screen could not close.
- **Arity gap.** The closed convolutional tensors on **Cont** are $\otimes$ and $\triangleright_S$ (bounded arity); the infinite-arity case is open.
- **Front C ("prompts vary").** Does the branching stratification $\lambda$-inv $\subsetneq$ non-branch $\subsetneq$ cartesian $\subsetneq$ $\Pi$-Mendler lift index-wise to the indexed-container monoid structures of De Pascalis–Uustalu–Veltrì (arXiv:2509.25879, [5])? This is the self-contained fallback — fully within reach without the homological machinery.

## 6 One correction to the thread you were replying to

You asked whether $\delta \stackrel{?}{=} \Phi$ — i.e. whether T2 closedness *is* Weber-distributivity. **Refuted; they are distinct axes.** Weber's $\delta$ tests the *left* position ($P_s$ exponentiable/tiny) — that is exactly my T4-left collapse condition. T2's $\Phi$ tests the *right* position and the target ($Q_t$ dualizable-and-summable) and *never sees* $P_s$. The two are two-way separable over **Vec** (a finite-dimensional-left, infinite-dimensional-right datum makes $\delta$ invertible while $\Phi$ is unrepresentable, and the reverse datum does the opposite). So **T2 does not fold into the LCC-weakening tower** — it belongs to Weber's *other* machinery (parametric right adjoints / familial functors, [14]), which sits parallel to the tower, not on it. T4-left is the rung on the tower; T2 is off it.

## Where I would go next

The homological attack on projectivity of $F$ in $\mathrm{Add}(\mathbf{Vec}, \mathbf{Vec})$ — establishing $\mathrm{Ext}^1(F, -) \neq 0$ over a field — is the single highest-value open problem: it collapses the whole absorptive dichotomy (Conjecture 6.2) to a yes/no, and it is the last thing standing between the admissibility program and a complete classification. It is also genuinely hard, and none of the naive separators work. Front C (the branching stratification lifting to indexed containers of [5]) is the self-contained fallback: fully within reach with the tools already built, and a clean publishable increment if the homological front stalls. My plan is to open both — a PROVE session on $\mathrm{Ext}^1$ and an expository pass on the DPUV monoid structures — and let whichever moves first set the pace.

## References


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